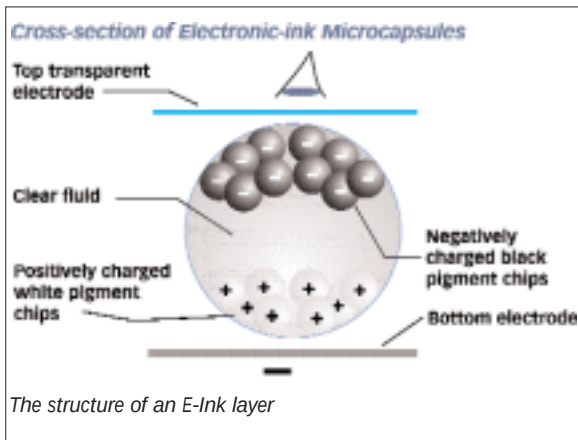




dred years. To achieve a device with the familiarity of paper, blended with the convenience of digital technology is the ultimate goal for portable display systems. Look around you right now, and see the amount of paper we still consume, including this magazine you are holding. Most of it could very well be viewed as electronic paper, a device that reads, looks and feels like paper, but shows you Web pages or MS Word documents or the pages of an electronic book.

There are two key criteria that any electronic paper display must satisfy—have the readability and portability of paper, and consume very low or no power when in use. That's right, no power when it is being read or looked at!

E-Ink has been making waves for a while and they have now reached some level of success in creating high resolution displays. E-Ink's technology is quite simple in terms of the science behind it. The challenges are in fabrication of the ink layer itself. An E-Ink display consists of a thin film containing millions of tiny microcapsules layered over an electrical circuit. One half of the microcapsule contains negatively charged black pig-

ment and the other half contains positively charged white pigment. When a positive charge is applied below the capsule, it attracts the black pigment down and the white pigment is pushed up. The spot above the microcapsule now looks white. When a reverse charge is applied, the black pigments would be pushed up, while the white ones will be attracted down. When the charge is removed, the pig-

ment particles stay in whatever location they are. This means that once a display has been updated, it takes zero power to keep it that way. Contrast this with your CRT monitor that is refreshed more than 70 times every second, or an LCD that is a constant power drain because of the backlight.

LCD, OLED or CRT displays are transmissive, i.e., they emit light. An E-Ink display is reflective—it does not produce any light of its own, but relies on ambient light for readability, just like paper. The reflectivity of E-Ink materials is quite comparable to that of paper across a wide viewing angle. This means that not only is an E-Ink display readable inside a building, it gets brighter when you step outside in the sun, unlike LCD displays that become unreadable.

E-Ink has tied up with Philips to manufacture com-

plete displays. They provide the ink layer, while Philips provides the active matrix backplane and control electronics. Philips recently announced a new fabrication technique for the transistors of the active matrix backplane that works on a plastic sheet. This means that E-Ink displays are available as flexible sheets for a number of applications.

Future watch

There are a slew of exciting new display technologies on the horizon and all of them are optimised for different applications. We will, in all probability, have all three kinds of displays in our offices and homes, as and when they become affordable. Pervasive displays are very likely to be the leading edge of pervasive computing in our environments. Displays have to be accessible so that computers can be hidden away, in a dark cabinet somewhere in the home or office. With thin, light and conveniently foldable displays, we will truly have access to information and entertainment anywhere we chose. Add a lenticular sheet on top and it can all be real! 

SUMOD HAJELA

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LCD Fights Back

Cholesteric LCD displays are new form of display that brings major new areas of application within the reach of LCD technology. Regular LCD systems regulate the output of light from a backlight, much like a fader switch. Cholesteric displays reflect light from a sidelight instead, and the display control is by varying their levels of reflectivity. To top it off, they are bi-stable, i.e., they can remain in either on, or off state without any power required to maintain it. Kent Display systems is a leading technology patent holder, and they have licensed it to about half a dozen manufacturers. Both Philips and Samsung SDI have demonstrated their own Cholesteric LCD displays. Samsung expects to be selling the foldable version for E-Book reader applications.